

KUSAL DEY

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PROFESSIONAL SUMMARY

Information Technology undergraduate (B.Tech, expected 2027) with a strong foundation in full-stack web development, data structures, and object-oriented programming. Skilled in Java, Python, and C, with hands-on experience building scalable, AI-driven applications using Next.js, Node.js, and FastAPI. Eager to contribute analytical thinking, clean coding practices, and a problem-solving mindset to an entry-level Software Developer role.

TECHNICAL SKILLS

Programming Languages: C, Java, Python

Web Technologies: HTML, CSS, REST APIs

Frameworks & Libraries: Next.js, Node.js, FastAPI, Tailwind CSS

Databases: Supabase, SQL

Tools & Platforms: Git, GitHub, VS Code, Docker, Vercel

Core Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS

KEY SKILLS

Problem-Solving • Analytical Thinking • Quick Learner • Self-Motivated • Team Collaboration • Adaptability

EDUCATION

B.Tech in Information Technology

July 2023 – Present

Narula Institute of Technology — CGPA: 8.94 / 10 (up to 6th semester)

Higher Secondary (Class XII), CBSE

2021 – 2022

Army Public School, Barrackpore — Score: 80.80%

Secondary (Class X), CBSE

2019 – 2020

Army Public School, Barrackpore — Score: 91.60%

PROJECTS

VectorDB Engine — Custom AI Search & RAG Pipeline | [Github](#)

Jan 2026 – Mar 2026

Tech Stack: Python, FastAPI, NumPy, Docker, Ollama, HNSW, RAG

- Architected a secure, containerized vector database from scratch using FastAPI and NumPy, implementing custom HNSW indexing and cosine similarity.
- Achieved sub-10 ms semantic search latency on a 200-record dataset, validating retrieval efficiency.
- Engineered a hybrid search algorithm and integrated local LLMs via Ollama to power a fully offline, Docker-deployed Retrieval-Augmented Generation (RAG) pipeline.

StockEasy — Cloud-Native Pharmacy Management SaaS | [Github](#) | [Live](#)

May 2026 – July 2026

Tech Stack: Next.js (App Router), TypeScript, PostgreSQL (Supabase), Upstash Redis, QStash, Gemini AI, Tailwind CSS

- Architected a secure, multi-tenant SaaS platform using Next.js and Supabase, implementing PostgreSQL Row Level Security (RLS) and Upstash Redis edge caching for strict tenant isolation.
- Engineered a custom FEFO (First-Expire-First-Out) algorithm for the POS system to dynamically prioritize near-expiry inventory, actively reducing dead-stock financial waste.
- Integrated a Gemini-powered AI analytics assistant and offloaded serverless background events using QStash.

CERTIFICATIONS

- SAP Certified - Back-End Developer - ABAP Cloud
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- NPTEL — Programming in Java
- NPTEL — Introduction to Machine Learning

LANGUAGES

English (Fluent) • Hindi (Fluent) • Bengali (Native)

ACHIEVEMENTS

- Strong academic record: CGPA 8.94 / 10 in B.Tech (IT) and 91.60% in CBSE Class X board examinations.
- Earned Elite NPTEL Certification in Machine Learning, ranking in the Top 1% of certified candidates.
- Built and deployed two production-style projects: an offline RAG pipeline and a multi-tenant pharmacy SaaS.